A replication and formalization of the three-trait QTL SEM in Mi (2009) and Mi, Eskridge, Wang, and coauthors (2010)

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Abstract

Can the errors in a multi-trait quantitative-trait-locus model stay correlated, after environment and block effects are removed, and still be recovered from the covariance? Mi, Eskridge, Wang, and coauthors (2010) fit a structural equation model that lets some traits influence others. In their simulation the model detects three loci more often than least squares: powers of 64%, 77%, and 96%, against 57%, 69%, and 94%. We proved which error models on their printed three-trait pattern are recoverable, and we reran the simulation. Uncorrelated errors determine the coefficients. One extra error correlation does not. Dropping one direct path and freeing that correlation does, and the estimator is consistent. Reading the minus signs in equation (7) by measuring their positions on the page reproduces Table 2, including the total effect 0.75. The same reading does not reproduce Table 1. Our detection rates are 50.8%, 53.5%, and 86.1%.

1 What this replication did

This section answers what we contributed, as distinct from what Mi et al. published. We contributed finished proofs, a set of simulations, and the comparison in Table 1. Their contribution is the model, the simulation design, and Tables 1 and 2. Commands are in Appendix B.

1.1 The Lean formalization

The proofs say which error models on the printed pattern can be recovered from the covariance, and which estimators converge. They are finished. Nothing is left as a placeholder.

Lean is the proof assistant. The sources are in `formalization/correlated-residuals/CorrelatedResiduals/`. The toolchain is Lean 4.22.0 and Mathlib v4.22.0, built with `lake build`. The only axioms are `propext`, `Classical.choice`, and `Quot.sound`. The first is propositional extensionality. The second is the axiom of choice, used for the matrix inverse and for positive-definiteness. The third is the construction of the real numbers. *(our own Lean proof)*

The statements, in plain words, are:

- `recover_sigma` (`Diagonal.lean`:246). If the trait errors are uncorrelated, the coefficients are a function of the covariance. Ψ is the error covariance. Φ is the covariance of the locus genotypes. Σ is the covariance of all observed variables.
- `bow12_same_law` (`Collision.lean`:114), and the same argument for the other two pairs. Adding exactly one error correlation makes the model non-unique: several parameter values produce the same covariance.

- `recover_bow` (`BowFree.lean`:187), `recover_bow01` (`BowMissing.lean`:148), and `recover_bow23` (`BowMissing.lean`:353). A bow-free pattern drops one direct path and frees the error correlation of that pair, following Brito and Pearl (2002). Each of the three such patterns determines the coefficients.
- `unitUpper_factor_unique` (`Triangular.lean`:68) and `sigmaRec_injective` (`Triangular.lean`:298). The same uniqueness holds in every dimension, not only for three traits.
- `plug_in_consistent_ae` (`Law.lean`:174) and `plug_in_bow_consistent_ae` (`BowConsistency.lean`:270). The estimator that plugs the sample covariance into the recovery formula converges almost surely. This is not a central limit theorem.

The Gaussian standard-error formula is a derivative plus a numerical check in `delta_method.py`. It is not one of the Lean proofs. The proof text is in Appendix A, taken from commit `05d69ea14 1adf3d5ee936d3b48061bde241250dc`.

1.2 The simulation studies

The simulations ask whether the published power advantage appears under the design the 2010 paper states. They are Monte Carlo experiments, not proofs.

Three studies are in `simulation/studies.py`. Study 1 is a 100-replicate scan that places one minus sign in front of the whole coefficient matrix. That placement was a misreading, kept so the later correction can be compared with it. Study 2 fits the bow-free model of Section 1: 400 replicates, seed `20101028`. Forcing the errors to be uncorrelated estimates the missing path as 0.8ρ, where ρ is the correlation that was actually present. The bow-free estimator covers the true coefficients at about the nominal 95% rate (about 0.915 to 0.958). Study 3 only checks scale. It refits the paths to three correlations from Campbell et al. (2003) and is not a replication of Table 1. *(our own simulation)*

Further runs change one choice at a time (`simulation/gaps.py` and `simulation/assumptions.py`) or move the first locus and the sample size (`qtl1_search.py`). Each run records whether the choice is one the simulation paragraph actually states.

1.3 The replication results

Table 1 is the comparison. A structural equation model (SEM) is the model with paths between traits. Least squares (LS) is the multi-trait regression with those paths removed. A single-trait scan fits one trait at a time. The position reading is the assignment of minus signs obtained by measuring where they sit on the page. Our powers under that reading use 150 lines, seed `20101101`, and 1000 replicates.

Table 1 reports SEM powers 64, 77, and 96. We found 50.8, 53.5, and 86.1. The verdict is no match. Section 4 shows that no design consistent with the simulation paragraph closes all three gaps at once.

Table 1 reports LS powers 57, 69, and 94. We found the same rates as for the SEM, because the two likelihood ratios are the same statistic on this model. The verdict is no match. Section 4.5 explains the discrepancy.

Table 1 reports single-trait powers 6, 3, and 97 on the first trait. We found 3.2, 1.2, and 88.6. The verdict is a partial match: the first two cells are close, and the third is not. Placing the minus sign in front of the whole matrix had given 13% for that third cell. Measuring the sign positions raises it to 86.2% (seed `20101041`) and does not reach 97.

Table 1: Published claims and our calculations. Powers are percentages for the three loci, in order. The position reading is defined in the text.

Claim	Published	Ours	Verdict
Table 1, SEM power	64, 77, 96	50.8, 53.5, 86.1	No match
Table 1, LS power	57, 69, 94	50.8, 53.5, 86.1	No match
Table 1, single-trait power, first trait	6, 3, 97	3.2, 1.2, 88.6	Partial
Table 2, total effect of locus 3 on trait 1	0.75	0.75	Match
Table 2, two locus-2 effects	−0.435, 0.065	−0.4375, 0.0625	Close

Table 2 reports a total effect 0.75 of the third locus on the first trait. We found 3/4. The verdict is a match. The matrix is

$$(I - M)^{-1}G = \begin{pmatrix} 1/8 & 0 & 3/4 \\ 3/8 & -7/16 & 3/8 \\ 1/4 & -1/8 & 1/4 \end{pmatrix}.$$

Section 3 records the character positions that fix the signs.

Table 2 reports −0.435 and 0.065 for two effects of the second locus. We found $-7/16 = -0.4375$ and $1/16 = 0.0625$. The verdict is close. The difference is small enough to be a rounding or transcription discrepancy. Ordinary three-decimal rounding of our values does not produce the printed decimals, so that reading is not settled. Section 4.4 keeps the graded account. The same two printed decimals appear in the 2010 *Genetics Research* companion.

No run that keeps the printed positions, the printed sample size of 150, and the printed threshold reproduces 64, 77, and 96 together. The nearest triple is 65.2, 73.3, and 99.3, from a first locus at 24 cM and 250 lines. That design is not the design in the simulation sentence.

2 Assumptions

This section answers which choices are ours rather than the paper's. The answer is the five subsections below. A choice the simulation paragraph states is not listed. Section 12 turns the choices off one at a time.

2.1 Design

We had to supply the breeding scheme and the map function, because the 2010 simulation paragraph does not. The population is an F1 selfed for twelve generations, with residual heterozygotes coded ±1 at equal probability. Mi et al. (2010) say recombinant inbred lines and do not give the generation count or a segregation-distortion model. The map function in the SAGMB scan is Haldane. The 2008 conference note names Haldane and prints the infinite-RIL expansion $2r/(1+2r)$; the SAGMB simulation paragraph does not. Cofactors are every marker except the flanking pair, which is the sentence on PDF page 10. The conditional expectation of the QTL genotype given the flanking markers follows Haley and Knott (1992), as the methods section says, implemented as the exact two-point expectation on the twelve-generation pedigree. *(assumption, flagged)*

2.2 Map positions

Marker coordinates and the detection window are ours. The locus positions are the paper's. Markers are placed at $0, 10, \ldots, 100$ cM. The sentence says eleven evenly spaced markers on a 100 cM chromosome. Putting a marker at each end is a reading of "evenly spaced," not a coordinate the paper prints. QTLs are placed at 22, 42, and 78 cM, which the sentence does print. The scan step is 1 cM. A detection is counted in a window of ± 10 cM around the true position, and the position standard deviations use that window. The paper's position standard deviations are consistent with a window nearer ± 5 cM, which the text does not state. The ± 10 cM window is ours. *(assumption, flagged)*

2.3 Sample size

The comparison in Table 1 uses more replicates than Table 1, and the same number of lines. The SAGMB simulation uses 150 lines and 100 replicates. The baseline powers quoted above use 1000 replicates of that 150-line design, so that Monte Carlo error is smaller than the gap under discussion. The 100-replicate size is what Table 1 itself is. Companion designs that were simulated and then set aside are 250 lines (Mi et al., Genetics Research, 2010) and 300 lines (the 2008 conference note). Neither is the SAGMB sample size. *(assumption, flagged where the replicate count exceeds the printed* 100*)*

2.4 Thresholds

The baseline uses the two critical values the paper prints. A multi-trait detection uses $\chi^2_{0.005,4} = 14.86$. A single-trait detection uses $\chi^2_{0.005,2} = 10.60$. Both numbers are printed. Treating 14.86 as a LOD score, or as a χ^2 critical value on 3 or on 1 degree of freedom, was tested and is not an assumption of the baseline. The permutation threshold, a 95% quantile near 16.1 under the baseline population, was computed for comparison and is not the threshold in the scan that produced 50.8, 53.5, and 86.1. *(the printed thresholds, used as printed)*

2.5 Readings of equation (7)

Two readings of the same display were programmed. Only the second matches Table 2.

The earlier reading puts the displayed minus in front of M and takes every entry of G as positive, so $B = -M$. Here B is the matrix of paths among traits, M is the matrix as printed, and G holds the locus effects. The total effect of QTL3 on Y_1 is then $1/4$, and the single-trait power for that cell is 13% at 100 replicates. QTL means quantitative trait locus. Y_1 is the first trait. This reading is study 1.

The position reading measures where each minus sign sits on the page. It is $y = My + Gx + \varepsilon$, with

$$M = \begin{pmatrix} 0 & 1/2 & 1/4 \\ 0 & 0 & -1/2 \\ 0 & 0 & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} -1/8 & 1/4 & 1/2 \\ 1/2 & -1/2 & 1/2 \\ 1/4 & -1/8 & 1/4 \end{pmatrix},$$

so $B = M$, and $\Psi = \mathrm{diag}(1.6, 1.8, 2.5)$ as printed in equation (8). The baseline, the joint search, and the Table 2 arithmetic use this reading. The next section says how the signs were assigned. *(our reading of the measured sign positions)*

3 Ambiguities that were resolved

This section answers which ambiguities in the printed equation we could settle, and how. The answer is the sign placement. It reproduces Table 2's 0.75 and removes most of the single-trait gap on the third locus. It does not produce the SEM powers in Table 1.

3.1 The sign in front of M

Putting one minus sign in front of M does not reproduce Table 2. The first programs read equation (7) as $y = -My + Gx + \varepsilon$. That population does not produce Table 2's total effect 0.75, and it produces a single-trait power of 13% where Table 1 prints 97% for QTL3 on Y_1.

The PDF content stream on page 10 of the ARS copy of Mi et al. (2010) settles the operator. The equals sign ends at $x = 145.9$ points. The parenthesis of M starts at $x = 148.1$. No minus sits between them. Four unary minuses are present, and each ends on the first digit of the number it signs: $G_{11} = -0.125$ (the minus ends at $x = 250.7$), $M_{23} = -0.5$ (ends at $x = 194.5$), $G_{22} = -0.5$ (ends at $x = 316.7$), and $G_{32} = -0.125$ (ends at $x = 316.7$). *(our reading of the measured sign positions)*

The measurement is stored in `simulation/out/qtl1/glyphs.json`, committed in `b4e8a3c594f12f419560d5cb2a569b649ae93a55`. The command that re-reads the same page and checks those coordinates is

```
python3formalization/correlated-residuals/simulation/measure_glyphs.py--
pdflocal-extraction/literature/unl-sem/mi-qtl2010.pdf--checkformalizatio
n/correlated-residuals/simulation/out/qtl1/glyphs.json
```

The PDF is not in the repository. The copy used here has SHA-256 `27c4f3926c9b5af9b399c9b3ef36918c28e7967ef777444b620279dd91628a81`.

3.2 Table 2, including 0.75

The position reading matches Table 2's total effect of 0.75. Two other sign placements do not. With $B = M$ and the signed G from the position reading, $(I - B)^{-1}G$ has $(1, 3)$ entry $3/4$. The rest of the parameter block matches Table 2 except the two QTL2 cells discussed in Section 4.4. The command is

```
python3formalization/correlated-residuals/simulation/table2_totals.py
```

It writes `simulation/out/table2_totals.json` and exits with status 1 if the $(1, 3)$ entry is not $3/4$ or the QTL2 middle total is not $-7/16$. *(our own calculation)*

The earlier reading $B = -M$ with unsigned G gives $1/4$ in that cell. An intermediate reading, $B = +M$ with unsigned G, gives $7/8$. Neither is 0.75.

3.3 The single-trait power on the earlier reading

The early 13% result was the sign error, not the published design. The 13% figure is the single-trait detection rate for QTL3 on Y_1 under $B = -M$ and unsigned G, 100 replicates, seed `20101027`. It is recomputed by `simulation/gaps.py`, which writes `simulation/out/gaps.json` and `simulation/out/gaps_table.csv`, committed in `c2769042e51806a77b2b70237fb4c014552e78fb`.

```
cd formalization/correlated-residuals/simulation
python3 gaps.py
```

Under the position reading the same cell is 86.2% at 1000 replicates, seed `20101041`, in that file. The printed value is 97. Measuring the sign positions accounts for most of the difference between 13 and 97. It does not meet 97. *(our own simulation)*

3.4 Where the 2008 counts sit

The 2008 conference counts are a different experiment, so they do not relabel Table 1. The 2008 conference paper, DOI `10.4148/2475-7772.1095`, SHA-256 `ab2cc7ddc94025609cdaa83a6353b22dcdbb4693fedf2b70df71f5fc8aa1452a`, is a different design: 300 lines and one QTL at 42 cM. Its Table 2 text layer gives multi-trait SEM overall power 91, multi-trait least-squares overall power 90, and single-trait powers 2, 39, 4, and overall 42. Trait-specific cells of the two multi-trait rows were not in the text layer. Those counts are not a relabelling of SAGMB Table 1. The file is recorded in `SOURCES.md`. Our three-QTL scan at 300 lines, which is not the conference design, gave SEM powers 89.3, 85.0, and 99.9. *(taken from a cited source, for the conference counts)*

4 What remains unresolved

This section answers what still disagrees with the published tables after the sign positions are fixed. The answer is the SEM power triple, two decimals in Table 2, and the SEM-versus-LS gap. A difference between a printed number and a number computed here is a discrepancy under our reading, or a gap this search did not close. It is not, by itself, evidence that the printed number is a mistake.

4.1 The gap between 64/77/96 and 51/54/86

Table 1's SEM powers do not match the printed design once the signs are read from the page. Table 1 prints SEM powers 64, 77, and 96 for the three QTLs. The position-reading baseline at the printed positions, $n = 150$, threshold 14.86, and 1000 replicates gives 50.8, 53.5, and 86.1 (`simulation/out/assumptions/baseline.json`, seed `20101101`, commit `82ff59d6bcfb02c273aaf1f83195e0d120fe0915`). Rounded to the nearest percent, the comparison is 64/77/96 against 51/54/86. The command is

```
cd formalization/correlated-residuals/simulation
python3 assumptions.py --configs baseline --reps 1000
```

The absolute gaps are about 13, 24, and 10 points. QTL2 is the large one. A second seed stays in the same range. Runs that change one modeling choice at a time are in Section 12. None of the choices the simulation paragraph leaves open closes the QTL2 gap while keeping the other two cells near the printed triple. *(our own simulation, set next to a cited table)*

Two calculations reproduce the middle cell and miss the first. At $n = 250$ and the printed positions the SEM powers are 81.6, 77.1, and 98.7. The analytic approximation with the other QTL genotypes controlled, at $n = 150$, is about 81, 77, and 100. Both put QTL1 near 80 rather than near 64. Replacing Ψ by I gives about 84, 80, and 99, which is the same shape and contradicts equation (8). An oracle scan that uses the realized genotype gives about 65, 64, and 94. A χ^2 threshold on 3 degrees of freedom, which is not the printed 14.86 on 4 degrees of freedom, gives 62.6, 64.6, and 92.9 on the baseline.

4.2 No text-consistent design matched all three numbers

No design the simulation paragraph states produces 64, 77, and 96 together. The search program varies the first QTL position, the sample size, a marker gap around QTL1, local distortion, residual heterozygosity, a two-plot line mean, and two cofactor rules that the simulation sentence does not describe. Each specification is tagged with whether it is consistent with the printed sentence. The search did not find a specification that is consistent with that sentence and close to all three of 64, 77, and 96.

The smallest sum of squared errors among the percentage triples computed in that search is the inconsistent point QTL1 at 24 cM and $n = 250$: SEM powers 65.2, 73.3, and 99.3, squared error about 26, of which the third cell is the larger share. The output is `simulation/out/qtl1/qtl1_24_n250.json`, commit `b4e8a3c594f12f419560d5cb2a569b649ae93a55`.

```
cd formalization/correlated-residuals/simulation
python3 qtl1_search.py --configs qtl1_24_n250 --reps 1000
```

Moving QTL1 off 22 cM pulls the first power toward 64 and pulls the second down from 77. The 99.3 against a printed 96, read as a Binomial$(100, 0.993)$ count, has lower-tail probability about 0.006 for a count of 96 or fewer. That comparison is descriptive. It does not identify a mechanism.

A global-maximum reading, one peak on the chromosome, produces powers on the order of 16, 14, and 65. The printed powers sum to 237, so a rule that returns only one detection cannot be the source of all three printed numbers. Reading 14.86 as a LOD threshold produces power 0 at these effect sizes.

4.3 A probability on the order of 10^{-12}

The published triple is far from our rates, on a rough probability scale. Table 1 is a 100-replicate experiment. Taking the baseline rates 0.508, 0.535, and 0.861 as the success probabilities,

```
python3formalization/correlated-residuals/simulation/printed_probability
.py
```

writes `simulation/out/printed_probability.json`. The output is the product of three binomial probability mass functions, $\mathrm{Bin}(100, p_i)$ at 64, 77, and 96. The product is 1.37×10^{-12}.

The product treats the three QTL counts as independent. They are computed from the same replicates, so they are not independent. The figure is an order of magnitude under that extra assumption. It says that a 100-replicate draw from the baseline rates is a poor match to the printed triple. It does not say how the printed triple was obtained. *(our own calculation, under an independence assumption we flag)*

4.4 The printed pair -0.435 and 0.065

These two cells are close to our arithmetic and are not a simple rounding of it. Table 2, and Table 3 of the Genetics Research companion (the ARS file Man851), print -0.435 and 0.065 for the QTL2 effects that the position reading gives as $-7/16 = -0.4375$ and $1/16 = 0.0625$. The same two decimals appear in both papers we could read. Rounding -0.4375 to three decimals is -0.438, not -0.435, and rounding 0.0625 to three decimals is 0.063 or 0.062 according to the tie rule, not 0.065.

This is a source discrepancy under our reading of which cells those printed numbers occupy. It is consistent with a rounding of a different intermediate quantity, with a typesetting substitution,

or with a coefficient we have not reconstructed. The two papers agreeing with each other makes a one-time typographical slip a weaker candidate, and it does not choose among the others. *(source discrepancy (our reading))*

4.5 SEM power above least-squares power

Table 1 reports higher SEM power than LS power on a model where the two tests are the same statistic. On the recursive model the multi-trait SEM likelihood ratio and the multi-trait least-squares likelihood ratio are the same statistic. In the baseline profiles the maximum absolute difference of the mean profiles is about 7×10^{-14}. Table 1 nevertheless prints SEM powers 7, 8, and 2 points above the least-squares powers (64 against 57, 77 against 69, 96 against 94). Figure 2 does not print a numeric scale that would let those gaps be read off the plot.

Equal likelihood ratios make equal powers under a common threshold. A difference of that size, repeated at all three QTLs, is a source discrepancy under our reading of the two columns. It would also arise if the two columns were computed under different thresholds or different cofactor sets. The simulation paragraph describes one scan. The present runs do not separate those possibilities. *(source discrepancy (our reading))*

5 Code and source availability

This section answers what source material we could and could not obtain. We obtained the 2008 paper, the 2010 author manuscript, and the *Genetics Research* companion. We did not obtain the dissertation, the publisher's version, or the programs. Each item was checked.

- The 2008 conference paper (DOI `10.4148/2475-7772.1095`) contains no code and no code supplement. The software sentence names SAS or Mplus. SHA-256 of the PDF we read: `ab2cc7ddc94025609cdaa83a6353b22dcdbb4693fedf2b70df71f5fc8aa1452a`.

- The De Gruyter page for SAGMB article 38 (DOI `10.2202/1544-6115.1552`) showed no supplement, and the version of record is paywalled. The file replicated here is the ARS manuscript Man850, SHA-256 `27c4f3926c9b5af9b399c9b3ef36918c28e7967ef777444b620279dd91628a81`. The publisher file was not compared with that manuscript. The request for the version of record is https://github.com/alectionary/vibe-math/issues/19.

- Mi's dissertation, *Genetic applications using structural equation modeling*, University of Nebraska–Lincoln, 2009, is restricted. The catalog page is https://digitalcommons.unl.edu/dissertations/AAI3350451/ and the ProQuest record is document `304940336`. The full text was not obtained. The acquisition request is https://github.com/alectionary/vibe-math/issues/18.

- The Genetics Research companion (ARS file Man851, SHA-256 `630a5e7b90665fd680d58915f6f9fe45e5051e8a5f6fff6af2cf16f8ae3b3739`) says that the programs were written in SAS PROC IML and are available by email to `xjmixu@yahoo.com`. No message was sent. The request for those programs, without sending the email, is https://github.com/alectionary/vibe-math/issues/20.

- Xiaojuan Mi works in industry as a contractor for Eli Lilly via Tech Data Services, and was formerly at Duke Clinical Research Institute. No academic webpage was found. Kent Eskridge's faculty page was not used as a substitute for the code or for the dissertation.

6 Next steps

This section answers what would resolve the open gaps, and what this note does not do. Three steps are open.

The dissertation is the document most likely to contain the SAS code, the marker list, and any appendix the articles omit. The route is an interlibrary loan or another institutional copy. Filing that request is outside this replication. The queue entry is issue 18, cited above.

A letter to Kent Eskridge, and a letter to Xiaojuan Mi, could ask for the PROC IML programs and for the design details that decide among the readings in Section 4. Those letters are to be drafted and sent by a person. This replication does not send email and does not write the letters.

Work that does not need those documents is local. The wheat fit is still blocked on the line identifiers, the analysis-of-variance residuals, the ± 1 genotype codes, and the edge set of Figure 5. Three further identification questions are open and are not theorems here: which mixed graphs have generically recoverable coefficients (Drton, Foygel, and Sullivant, 2011); an infinite family of models that are rationally recoverable but fail the half-trek test (Foygel, Draisma, and Drton, 2012); and whether the ancestral decomposition of Drton and Weihs can be done in polynomial time. A Lean proof of the delta-method central limit theorem is the natural next statement about the estimator already proved consistent.

7 The question, as they pose it

This section quotes the question the proofs address. The authors ask whether trait errors can be correlated after environment and block effects are removed. The source is Mi et al., Statistical Applications in Genetics and Molecular Biology 9 (2010), article 38. On printed page 18 (PDF page 20) they write:

> Here, we assumed that the residual errors of the traits were independent which we believe is a reasonable assumption since we removed the effects of environment by using, for the y values in equation (2), the residuals from a main effects linear model with environment and block within environment as the main effects. However, even after removing environment main effects, the traits may still be correlated and the trait error terms can be modeled as being correlated with each other.

(taken from a cited source) The parent request is that extension. It is not a numbered conjecture. Mi's 2009 dissertation (ProQuest AAI3350451) is the obvious place a fuller design might have been written. It is restricted to a ProQuest preview. It was not consulted.

8 The restriction

This section says which model the proofs cover. They cover the three-trait pattern in equation (7), not the four-trait wheat graph in Figure 5. Equation (7) is the only fully printed structural pattern. On PDF page 10 the equals sign ends at $x = 145.9$ points and the parenthesis of M begins at $x = 148.1$. No operator fits between them, so the display is $y = My + Gx + \varepsilon$. The minus at $x = 189.1$ is inside M, on the middle row, and three further minuses sit on G_{11}, G_{22}, and G_{32}. *(our reading of the measured sign positions)*

$$M = \begin{pmatrix} 0 & 1/2 & 1/4 \\ 0 & 0 & -1/2 \\ 0 & 0 & 0 \end{pmatrix}, \qquad G = \begin{pmatrix} -1/8 & 1/4 & 1/2 \\ 1/2 & -1/2 & 1/2 \\ 1/4 & -1/8 & 1/4 \end{pmatrix}.$$

Equation (8) is $\Psi = \mathrm{diag}(1.6, 1.8, 2.5)$. *(taken from a cited source)* In $y = By + Gx + \varepsilon$ one then has $B = M$. A first reading placed the interior minus in front of M and took every entry of G as positive. That reading is the population of the first replication below. The zero pattern is the same either way: the free positions are still $(1, 2)$, $(1, 3)$, and $(2, 3)$. The wheat Figure 5 graph has four traits and is not this matrix. The restriction is covariance identification for this three-trait recursive pattern: diagonal Ψ, one extra residual covariance, and the three ways to drop one edge and free the residual pair that edge had used. Singular Ψ is outside the claim. The general mixed-graph characterization is not this restriction.

9 What we proved

This section answers the identification question in theorem form. On the printed pattern, uncorrelated errors are recoverable, one extra error correlation is not, and each bow-free pattern is recoverable and consistently estimable. The proofs are finished: Lean's `sorry` placeholder does not appear. The axioms are `propext` (propositional extensionality), `Classical.choice` (the axiom of choice), and `Quot.sound` (the construction of the real numbers). *(our own Lean proof)*

- `recover_sigma`: uncorrelated errors and a positive-definite genotype covariance determine every coefficient.
- `bow12_same_law` and its two siblings: one extra error correlation is not unique, for every matrix of locus effects.
- `recover_bow`, `recover_bow01`, `recover_bow23`: each of the three bow-free patterns determines the coefficients.
- `unitUpper_factor_unique` and `sigmaRec_injective`: the same uniqueness holds in every dimension.
- `plug_in_consistent_ae` and `plug_in_bow_consistent_ae`, with the two further bow-free patterns: independent samples with finite coordinate products make the plug-in estimator almost surely consistent.

Standard errors use a delta-method derivative and Isserlis' formula for the covariance of a sample covariance, checked in `delta_method.py`. *(our own written proof, and our own simulation)* That argument is not a Lean central limit theorem.

Statement	Lean declaration	File and lines
Uncorrelated errors	`recover_sigma`	`Diagonal.lean`, 246–279
One error correlation is not unique	`bow12_same_law` and two siblings	`Collision.lean`, 114–160
Bow-free recovery	`recover_bow`	`BowFree.lean`, 187–213
Other missing edges	`recover_bow01`, `recover_bow23`	`BowMissing.lean`, 148 and 353
Triangular factor	`unitUpper_factor_unique`	`Triangular.lean`, 68–135
Recursive identification	`sigmaRec_injective`	`Triangular.lean`, 298–317
Almost-sure consistency	`plug_in_consistent_ae` and `plug_in_bow_consistent_ae`	`Law.lean`, 174–199; `BowConsistency.lean`, 270–304

The links are pinned to commit `05d69ea141adf3d5ee936d3b48061bde241250dc`, which contains these Lean files. The simulation script and the JSON outputs are commit `5cd3c9f6d2f6e38f25e32146d74d55f3778436e1`. Appendix A prints the central proof bodies.

Brito and Pearl (Structural Equation Modeling, 2002) prove that a recursive linear model is identifiable when no residual covariance joins a variable to its direct effect. Our three bow-free left inverses are the three-trait case of that pattern, proved from the printed matrices rather than cited as a substitute for the algebra. Drton, Foygel, and Sullivant (Annals of Statistics, 2011) give a global criterion for mixed graphs; the half-trek criterion of Foygel, Draisma, and Drton (Annals of Statistics, 2012) is a generic sufficient condition. Neither is re-proved here, and the 2011 characterization stays a note with no inventory id. Li et al. (PLoS Genetics, 2006) and Valente, Rosa, de los Campos, Gianola, and Silva (Genetics, 2010) are earlier structural and recursive treatments of multiple traits. Hidalgo-Contreras, Salinas-Ruiz, Eskridge, and Baenziger (Agronomy, 2021) fit a later Nebraska wheat SEM; their lines are a 2013 F6 panel, not the 1999–2001 RICLs, and none of their estimates enter the simulations.

10 Replication design

This section turns the simulation paragraph into a program and lists what that paragraph does not say. The printed design, PDF pages 9–10, is 100 replicates of 150 recombinant inbred lines, one chromosome of 100 cM, 11 evenly spaced markers, additive QTLs at 22, 42, and 78 cM, phenotypes from equations (7)–(8), and a test every 1 cM. QTL genotypes are coded $+1$ and -1. Cofactors are all markers except the flanking markers. The printed critical values are 14.86 and 10.60. *(taken from a cited source)* Those two numbers are the chi-square 0.005 quantiles on 4 and 2 degrees of freedom (14.860 and 10.597). The paragraph on PDF page 9 writes the Bonferroni approximation $\chi^2_{\alpha/M,\,m+1}$. Ten marker intervals and $\alpha = 0.05$ give the tail probability 0.005, with $m + 1 = 4$ for three traits and $m + 1 = 2$ for one trait. *(our own calculation, from the cited formula)*

The paper does not state the following, so each is our assumption. *(our assumption)*

1. The map function is Haldane, with no interference.
2. An RIL is an F1 selfed for 12 generations. A leftover heterozygote is coded ± 1 with equal probability. There is no segregation distortion.
3. The eleven markers are at $0, 10, \ldots, 100$ cM.
4. The analysis uses the centered covariance with divisor n, which is the fitting function (6). We did not run SAS PROC TCALIS.
5. The flanking expectation is the conditional mean of the QTL genotype under that same 12-generation law. The 10 cM window mentioned later in the paper is for the wheat scan, not for this simulation, and was not used.
6. A named QTL is detected when the maximum likelihood ratio in $[0, 32]$, $(32, 60]$, or $(60, 100]$ exceeds the threshold. Effects are read at that argmax.

Seeds are `20101027` for this study. The outputs are `study1.json`, `study1_power.csv`, and `study1_effects.csv` under `simulation/out`.

For a complete recursive order, the Gaussian likelihood of the diagonal structural model equals the likelihood of the multivariate regression on the same exogenous regressors: the sum of the successive residual log variances is the log determinant of the residual covariance. The two joint likelihood ratios are therefore the same number. In the run, the maximum absolute difference of the two mean profiles is 7×10^{-14}. *(our own written proof, confirmed by our own simulation)* Mi et al. describe the two profiles as very close (PDF page 14). Their Table 1 powers still differ by 7, 8, and 2 replicates out of 100.

11 Replication results

This section gives the row-by-row account behind Table 1. The summary verdicts are no match for the two multi-trait power rows, a partial match for the single-trait row, a match for the total effect 0.75, and a close pair for -0.435 and 0.065.

Table 2 is an earlier scan, not the comparison in Table 1. It places the minus sign in front of the whole matrix. Our multi-trait powers keep the printed order, third QTL above the first two, and they sit 5 to 16 points below Table 1. The single-trait column for the first trait at the first QTL matches Table 1 exactly (6%). The single-trait column for the first trait at the third QTL does not: Table 1 prints 97% and the replication gives 13%.

Table 2: Detection percentages. "Mi" is Table 1 on PDF page 12. "Ours" is 100 replicates, seed `20101027`. SEM and LS coincide in our run.

QTL	SEM Mi	SEM ours	LS Mi	ST Y_1 Mi	ST Y_1 ours	ST overall ours
1	64	59	57	6	6	24
2	77	61	69	3	0	35
3	96	86	94	97	13	34

This first reading produces a total effect of 0.25, where Table 2 prints 0.75. The algebraic total effects under $B = -M$ are, by columns of G,

$$(I + M)^{-1}G = \begin{pmatrix} -1/8 & 0 & 1/4 \\ 3/8 & 7/16 & 3/8 \\ 1/4 & 1/8 & 1/4 \end{pmatrix}.$$

(our own calculation) Table 2's parameter block matches this matrix on Y_2 and Y_3 for the first and third QTLs, and it does not match it on Y_1. The printed total effect of the third QTL on Y_1 is 0.75, against 0.25 from this first reading. Section 12 tests that disagreement directly.

Direct-effect means at the SEM argmax are close to G where the signal is large. For the third QTL they are 0.460 (0.175), 0.476 (0.179), and 0.242 (0.190), against the printed SEM estimates 0.479 (0.176), 0.527 (0.166), and 0.257 (0.205) and against $G = (1/2, 1/2, 1/4)$. *(our own simulation, set next to a cited source)* Position means are 22.65 (7.28), 41.26 (5.25), and 79.85 (3.76), against the printed 21.57 (2.83), 41.95 (2.58), and 78.03 (2.16). The biases are small. Our standard deviations are larger. The mean profile exceeds 14.86 only at 40 cM and on 75–82 cM. Figure 2 is described as one simulated population, with excursions on 16–28, 38–48, and 70–88 cM. An average profile is a different object, and ours is lower.

12 Which hypothesis closes which gap

This section asks which single change moves our rates onto Table 1. The answer is that the measured sign positions fix the total effect 0.75 and most of the single-trait gap, and no text-consistent change fixes the SEM triple. Five changes were rerun from fixed seeds. The full grid is `simulation/out/gaps.json` and `gaps_table.csv`. Closure is $1-|\hat{p}-p_{\text{printed}}|/|p_{\text{first}}-p_{\text{printed}}|$, using the first reading's 100-replicate powers as the denominator. A value of 1 means the printed number is met. A negative value means the estimate moved away from the table. *(our own simulation)*

Sign convention. $B = +M$ with unsigned G gives total effect 0.875 on Y_1 for QTL3, not 0.75.

Table 3: Gap closure. SEM powers are the three QTLs. ST_3 is single-trait power for QTL3 on Y_1 (printed 97). ST_2 is single-trait power for QTL2 on Y_1 (printed 3). SDs are the three SEM position standard deviations. T_{31} is the algebraic total effect of QTL3 on Y_1 (printed 0.75).

Reading	SEM	ST_3	ST_2	position SD	T_{31}	ST_3 closed	T_{31} closed
Printed	64, 77, 96	97	3	2.83, 2.58, 2.16	0.75	—	—
First, $n = 100$	59, 61, 86	13	0	7.28, 5.25, 3.76	0.25	0	0
First, $n = 1000$	65, 63, 87	9	1	6.20, 5.32, 4.38	0.25	−0.05	0
$B = +M$, G unsigned	68, 57, 86	87	44	5.90, 5.33, 4.60	0.875	0.88	0.75
Position reading, $n = 100$	46, 45, 80	85	1	5.57, 5.93, 5.10	0.75	0.86	1
Position reading, $n = 1000$	52, 48, 87	86	1	5.15, 5.42, 3.47	0.75	0.87	1
Position reading, no cofactors	46, 65, 100	100	65	6.99, 9.12, 3.80	0.75	0.96	1
Position reading, wheat cofactors	34, 24, 89	95	1	7.33, 7.91, 3.77	0.75	0.98	1
Position reading, $\Psi = I$	80, 78, 98	97	2	4.86, 4.02, 2.79	0.75	1.00	1
Position reading, window ±5 cM	43, 41, 80	83	1	2.80, 2.76, 2.03	0.75	0.83	1
Position reading, $\alpha = 0.05$	77, 84, 98	95	14	5.57, 5.93, 5.10	0.75	0.98	1

The position reading gives

$$(I - M)^{-1}G = \begin{pmatrix} 1/8 & 0 & 3/4 \\ 3/8 & -7/16 & 3/8 \\ 1/4 & -1/8 & 1/4 \end{pmatrix}.$$

(our own calculation) Every Table 2 parameter cell matches this arithmetic except the QTL2 row for Y_2, where the exact values are $-7/16 = -0.4375$ and $1/16 = 0.0625$ and the table prints -0.435 and 0.065. *(source discrepancy (our reading))* The first reading's total of 0.25 is what produced the single-trait power 13%. Under the position reading, 1000 replicates (seed `20101041`) give 86.2% (Clopper–Pearson interval 83.9–88.3). That closes most of the 97 versus 13 gap and does not meet 97.

Scan method. The simulation paragraph on PDF page 10 fits every marker except the flanking pair, and the methods section replaces the QTL genotype by its flanking-marker expectation (Haley and Knott, 1992). Dropping the cofactors, on the position reading, raises the single-trait QTL3 power to 100% and the SEM power for QTL3 to 100%, and it leaves the first SEM power at 46. The wheat rule on PDF page 11, a $P < 0.1$ screen with at most five cofactors and a 10 cM window, gives single-trait power 95% (interval 88.7–98.4, which contains 97) and SEM powers 34, 24, and 89. Nearest-marker regression gives the same pattern as dropping cofactors. None of these is the sentence printed for the simulation, and none of them brings all three SEM powers onto 64, 77, and 96.

Threshold and window. One hundred phenotype shuffles of a position-reading replicate have 95% quantile 14.74 for the maximum likelihood ratio, next to the printed 14.86. A permutation threshold at $\alpha = 0.05$ does not move the powers. The 500 permutations in the paper are in the wheat section. Replacing 14.86 and 10.60 by the $\alpha = 0.05$ quantiles on 4 and 2 degrees of freedom (9.49 and 5.99) raises the position-reading SEM powers to 77, 84, and 98 and the single-trait QTL3 power to 95. That threshold is not the printed one. Taking the peak within 5 cM of the true site, under the position reading, gives position standard deviations 2.80, 2.76, and 2.03, against 2.83, 2.58, and 2.16. The midpoint windows used in the first replication were our assumption. The paper does not name the window that assigns a peak to a named QTL.

Unprinted design. Eleven evenly spaced markers on a segment of 100 cM, with markers at the ends, are at $0, 10, \ldots, 100$. Shifting them to $5, 15, \ldots, 95$ does not close the SEM gap. Standardizing each trait to sample standard deviation 1 leaves the likelihood ratio essentially unchanged. Replacing Ψ by I does close the single-trait QTL3 gap (97%) and puts the second and third SEM

powers on 78 and 98, while the first SEM power rises to 80 (interval 70.8–87.3, which does not contain 64). The simulation sentence says the residuals are drawn from matrix (8). $\Psi = I$ contradicts that sentence. It is a sensitivity, not the display.

Monte Carlo error. At 1000 replicates the first reading gives SEM powers 65.1, 62.5, and 86.6, with intervals 62.1–68.1, 59.4–65.5, and 84.3–88.7. The printed 64 lies in the first interval. The printed 77 and 96 do not. Single-trait power for QTL2 on Y_1 is 1.2% (interval 0.6–2.1) under that reading and 0.8% (0.3–1.6) under the position reading. The printed 3 is inside the $n = 100$ interval for zero detections (0–3.6) and outside the $n = 1000$ intervals. The discrepancy is three replicates in Table 1.

The most likely account is the position reading of equation (7), together with a peak window of about 5 cM that the paper does not state. That reading removes the 0.75 discrepancy and most of the single-trait QTL3 discrepancy, and the narrow window removes the position-SD discrepancy. It does not remove the SEM gap for QTL2 and QTL3. Those two percentages move onto the printed values only in runs that drop the printed cofactors or replace matrix (8). Both of those runs conflict with a sentence on PDF page 10.

13 Explanations considered

This section asks whether any choice the simulation paragraph leaves open closes the remaining power gap. None does. The cell 77 appears in two calculations that are not that paragraph, and both put the first locus near 80% rather than at 64%. The gaps that remained after the position reading, on a fresh 1000-replicate run (seed `20101101`), are SEM powers 50.8, 53.5 and 86.1 against 64, 77 and 96, single-trait power for QTL3 on Y_1 of 88.6 against 97, and single-trait power for QTL2 on Y_1 of 1.2 against 3. The intervals are 50.4–56.6, 83.8–88.2, 86.5–90.5 and 0.6–2.1. A second seed (`20101102`) gives SEM powers 49.4, 48.3 and 84.6. The QTL2 gap of about 24 points is not a seed effect. In the table, group A changes one modeling choice at a time. Group B compares the printed counts with the arithmetic, with a noncentral chi-square approximation, and with the companion designs. Group C changes the sample size. The grid is `simulation/out/assumptions/`. *(our own simulation)*

No choice the simulation paragraph leaves open closes the QTL2 gap. The infinite-RIL formula is the one printed in the 2008 conference note, and Kosambi is unnamed in SAGMB; together they move QTL2 from 53.5 to 54.2. The no-double-crossover expectation in that note, and an F2 expectation applied to RIL data, stay inside the seed range of the baseline. A 95% permutation threshold is 16.1, and the median of 40 per-replicate thresholds, each the maximum of 19 null scans, is 16.9. Both are above 14.86, so a permutation rule makes the powers smaller (43.5, 44.2, 82.0 at 16.1). Replacing 14.86 by the $\alpha = 0.05$ quantile 9.49 raises the baseline to 84.7, 82.6 and 98.5, which is not the printed critical value. Standardizing the traits, scanning every 2 or 5 cM, and shifting the eleven markers off the ends do not close QTL2. The ± 1 and $0/2$ codes have the same homozygote contrast, so they have the same likelihood ratio; a $0/1$ code with the printed G halves that contrast and the power collapses. *(our own simulation)*

The printed 77 is consistent with two calculations that are not the printed marker scan. With the three QTL genotypes in each structural equation, the noncentral chi-square approximation at $n = 150$ and threshold 14.86 gives joint powers 81%, 77% and 100%. The partial single-trait power for QTL3 on Y_1 is 99.7%, and for QTL2 on Y_1 it is 0.1%. The same approximation at $n = 250$ and $n = 300$ is above 97% for every QTL. *(our own calculation)* The marker-cofactor scan at 250 lines, the sample size of Mi et al. Genetics Research (2010), gives SEM powers 81.6, 77.1 and 98.7, single-trait Y_1 powers 3.2, 2.2 and 99.0, and single-trait Y_2 powers 31, 38 and 32 against the printed

Table 4: Explanations considered. Closure is the fraction of the seed-20101101 gap removed. The four fractions are QTL2 SEM, QTL3 SEM, single-trait QTL3 on Y_1, and single-trait QTL2 on Y_1. The Family column is a group label: A changes one modeling choice, B compares the counts with a calculation or a companion design, and C changes the sample size.

Hypothesis	Family	Evidence	Gap closed	Verdict
Infinite RIL, $2r/(1+2r)$	A	SEM 56, 55, 91	.07/.54/.36/ − .06	open, and too small
Kosambi, infinite RIL	A	SEM 56, 54, 91	.03/.49/.31/ − .06	open in SAGMB; not Table 1
Flanking formula	A	within the seed range	about 0	cannot be distinguished from the 12-generation expectation
Per-equation tests	A	any equation above 10.60 gives 52, 52, 78	negative on QTL3	does not match
Permutation threshold	A	95% quantile 16.1 in 200 shuffles	power falls	14.86 is slightly liberal; the gap is not a low threshold
$\Psi = I$	A	SEM 84, 80, 99; $ST_3 = 98$	.89/.70/.87/.06	contradicts matrix (8)
Entries used as SDs	A	power falls to 24, 25, 59	negative	not the covariance sentence
Half contrast, 0/1	A	power falls to 11, 9, 23	negative	the sentence says +1 and −1
F2 or backcross	A	power falls	negative	the sentence says RIL
Oracle genotype	A	SEM 65, 64, 94; $ST_3 = 95$	.46/.85/.80/.94	not the flanking expectation
QTLs on markers	A	SEM 72, 73, 96	.84/.97/.85/.22	contradicts 22, 42, 78
No cofactors	A	ST_2 rises to 66	.59/.65/.70 and ST_2 breaks	contradicts the cofactor sentence
$n = 250$	B	SEM 82, 77, 99; Y_2 powers 31, 38, 32	1.00/.73/.76/.56	matches QTL2, not QTL1
$n = 300$	B	SEM 89, 85, 100	overshoots QTL2	above Table 1
Chi-square, QTLs controlled	B	powers 81, 77, 100 at $n = 150$	QTL2 matches	not the marker-cofactor scan

29, 43 and 33. QTL2's SEM cell matches. QTL1's SEM cell is 82 against 64. At 300 lines, the sample size of the 2008 conference note, the three-QTL scan is 89.3, 85.0 and 99.9, above Table 1. *(our own simulation)*

Those companion designs are not Table 1. Genetics Research simulates 250 lines and two QTLs, at 42 and 78 cM, with codes 2 and 0, and the detection rule is a posterior peak rather than the likelihood-ratio threshold. Its Table 2 powers are 99 and 100. Equation (15) and Figure 2 of that paper print the same path signs as the position reading, 1/2, 1/4 and −1/2, and the same rounded QTL2 row −0.435 and 0.065. *(taken from a cited source)* The conference note simulates 300 lines and one QTL at 42 cM, names Haldane's map and the expansion $2r/(1+2r)$, and uses codes 2 and 0. Its Table 2 text layer gives multi-trait SEM overall power 91, multi-trait least-squares overall power 90, and single-trait powers 2, 39, 4, and overall 42. Trait-specific cells of the two multi-trait rows were not in the text layer. Neither table is a relabeling of SAGMB Table 1. *(taken from a cited source)*

Table 2's QTL2 row prints −0.435 and 0.065 where the path product is $-7/16 = -0.4375$ and $1/16 = 0.0625$. The same printed decimals appear in the Genetics Research table. *(source discrepancy (our reading))* On a complete recursive model the multi-trait SEM and multi-trait LS likelihood ratios are the same number, and in every run here their powers agree. Table 1 prints SEM powers 7, 8 and 2 points above the LS powers, while the text says the two profiles are very close. *(source discrepancy (our reading))* Figure 2 is described as one simulated population and

does not print a numeric likelihood-ratio scale, so the power counts cannot be read off the figure.

The best account is that no open choice in the simulation paragraph produces Table 1, and that the cell 77 is consistent both with the chi-square power of these direct effects at 150 lines when the other QTL genotypes are controlled and with the marker scan at 250 lines. Both of those readings put the first QTL near 80% rather than at the printed 64%. The single-trait QTL2 cell, 3 against our 1.2, is consistent with the algebraic total effect of zero.

13.1 A joint reading of 64, 77 and 96

This subsection asks whether any one reading produces 64, 77, and 96 together. No reading that the simulation paragraph states does. Every entry of equation (7) was measured again on PDF page 10. The minus signs are four printed marks. One ends at $x = 250.7$ on $G_{11} = -1/8$. One ends at $x = 194.5$ on $M_{23} = -1/2$. Two end at $x = 316.7$, on $G_{22} = -1/2$ and $G_{32} = -1/8$. No other entry carries a minus, and there is no minus between the equals sign and M. The Table 1 digits are the characters 6 and 4, then 77, then 96. *(taken from a cited source)* The scans below use 1000 replicates. Squared error is the sum of squared differences of the three SEM percentages. The chi-square treats each printed percentage as a count out of 100 draws from the simulated rate. *(our own simulation)*

Table 5: SEM triples ranked by squared error against 64, 77, 96. The chi-square is for those three cells. A text-consistent row is one the simulation paragraph on PDF page 10 states. The Family column uses the same group labels as Table 4.

Hypothesis	Family	SEM powers	SSE	χ^2	Consistent with the text
QTL1 at 24 cM, 250 lines	A+C	65, 73, 99	26	16.4	no
QTL1 at 25 cM, 250 lines	A+C	62, 69, 99	74	13.8	no
QTL1 at 23 cM, 250 lines	A+C	72, 73, 99	80	8.6	no
Oracle genotype, 150 lines	A	65, 64, 94	163	7.4	no
Threshold $\chi^2_{3,0.995}$	B	63, 65, 93	165	8.3	no; printed value is 14.86
Marker gap around QTL1, 250 lines	A+C	78, 76, 98	212	13.6	no
Printed positions, 250 lines	C	82, 77, 99	317	26.3	no; the sentence says 150
Two plots averaged, Ψ at plot level	A	63, 59, 100	341	174	no
Local allele frequency 0.30 at QTL1	A	62, 61, 85	392	20.8	no
$\Psi = I$	A	84, 80, 99	416	39.3	no; contradicts (8)
Position-reading baseline, 150 lines	A	51, 54, 86	824	37.4	yes
One peak on the chromosome	B	16, 14, 65	4602	—	the printed sum is 237
14.86 applied to the LOD	B	0, 0, 0	—	—	14.86 is the chi-square value
G transposed	A	55, 94, 19	6308	441	no; columns are labeled QTL

The smallest squared error is 65.2, 73.3, 99.3. Single-trait powers on Y_1 are 2.6, 1.9, 99.1 against 6, 3, 97, and on Y_2 they are 22, 37, 33 against 29, 43, 33. QTL1's 65 cannot be distinguished from the printed 64 at the precision of 100 replicates. QTL2 remains about four points low. QTL3 at 99.3 makes a printed 96 unlikely if that rate were true: the lower-tail probability is about 0.006. The position 24 cM and the sample size 250 are not the simulation sentence. Moving QTL1 from 22 to 24 cM is what pulls the first power down from the 82 of the 250-line scan at the printed positions. The same move pulls the second power down from 77 to 73. The two cells do not land on 64 and 77 together. *(our own simulation)*

Family A does not supply a text-consistent joint match. Flipping the sign of G_{11}, which the remeasured minus sign does not support, gives 55, 50, 85. Halving G_{21} gives 17, 53, 87. A residual heterozygosity of 0.15 gives 20, 18, 57. Placing QTL1 10 cM from QTL2 gives 24, 22, 84. Kosambi

interference was already inside a few points of the baseline. Standardizing the traits was 51, 48, 87.

Family B does not supply the triple either. Stepwise cofactors and a two-stage refit lower the first two powers and raise the third, to about 33, 28, 98 and 29, 28, 98. A marker-bracket window, a 20 cM separation rule, and dropping the terminal 10 cM leave the baseline near 51, 54, 86. The χ^2_3 threshold 12.84 gives 63, 65, 93: the first cell matches, and the second does not. The printed critical value is 14.86, the χ^2_4 quantile. Reading 14.86 as a LOD multiplies the likelihood-ratio threshold by $2\log 10$ and the powers fall to zero. One peak per chromosome assigns the global maximum to a single QTL, so the three powers sum to at most 100. On the baseline they are 16, 14, 65. The printed powers sum to 237, so that counting rule cannot be the source of all three cells. The flanking regressor has positive variance at every interior test point on this marker grid, so a singular fit does not arise and counting non-convergence as a miss does not change the powers.

Family C. The probability of the printed counts 64, 77, 96 as three independent Bin(100) draws from the baseline rates 0.51, 0.54, 0.86 is about 10^{-12}. Swapping the SEM and LS columns of Table 1 produces 57, 69, 94, which is not a row of any scan here. The PDF prints the digit 6, not 8, so 64 is not an OCR reading of 84. The 250-line scan at the printed positions is 82, 77, 99. That matches the cell 77. It is near 84, 77, 96 in the first cell and not near 64. The Genetics Research powers are 99 and 100 under a different design. The 300-line three-QTL scan is 89, 85, 100.

The best joint account is negative. No tested reading that the simulation paragraph states produces 64, 77, and 96 together. The closest percentage triple uses a QTL position and a sample size that paragraph does not state, and its third cell is still high relative to 96. What remains is that gap: the first cell can be brought near 64 by weakening QTL1 or by a lower threshold, and the second cell can be brought to 77 by using 250 lines or by controlling the other QTL genotypes, and those two adjustments have not been found together in one design the text states.

14 Correlated residuals

This section answers what goes wrong if the errors are correlated and the fit still forces them to be independent. The uncorrelated fit invents a direct path. The bow-free fit, which drops that path and frees the error correlation, recovers the coefficients. The bow drops the edge into trait 0 from trait 2, sets that structural coefficient to zero, and frees the residual covariance $c = \rho\sqrt{1.6 \cdot 2.5}$. The other equation-(7) coefficients stay. The exogenous variables are the three QTL genotypes on the RIL map, not a marker scan. *(our assumption, matching* `recoverBow`*)* The sample is the centered second-moment matrix, $n = 150$, 400 replicates, seed `20101028`.

At the population covariance the bow plug-in recovers the parameter to machine precision. The diagonal plug-in does not. Its probability limit for the missing edge is 0.8ρ: zero at $\rho = 0$, 0.20 at $\rho = 0.25$, and 0.48 at $\rho = 0.6$. The QTL coefficient $G_{11} = 1/8$ has probability limit 0.075 at $\rho = 0.25$ and 0.005 at $\rho = 0.6$. The coefficient $G_{13} = 1/2$ moves to 0.45 and 0.38. *(our own simulation)* A Wald test that the missing edge is zero, using the diagonal delta-method standard error, rejects at rate 0.058 when $\rho = 0$ and at rate 0.728, 0.877, and 1 at $\rho = 0.25$, 0.30, and 0.60. The joint nine-degree Wald test of all QTL coefficients is only mildly inflated at $\rho = 0.25$ (rejection rate 0.065) and reaches 0.297 at $\rho = 0.6$. The damage shows up first as bias in two coefficients and as a spurious edge, not as an immediate explosion of the joint QTL test.

`recoverBow` is centered on the QTL and path coefficients. At $\rho = 0.25$ the Monte Carlo means of b_{12}, b_{23}, G_{11}, and G_{13} are -0.507, -0.498, 0.127, and 0.509. Coverage of the nominal 95% delta-method intervals, with the Gaussian Isserlis covariance plugged into the sample, is between 0.915 and 0.958 across the reported coefficients and the six correlations. The genotypes are ± 1, so

Isserlis is an approximation. *(our assumption inside the standard error)* The coverage near 95% says the approximation is tolerable at $n = 150$ for these coefficients. Variance-component means sit a little low (1.54 against 1.6 at $\rho = 0$) because the likelihood uses divisor n; the probability limit is still the population value.

The same seeds were rerun with the position-reading coefficients: $b_{12} = +1/2$, $b_{23} = -1/2$, the edge b_{13} removed from $+1/4$, and the signed G. *(our own simulation)* At $\rho = 0.25$ the diagonal probability limit of the missing edge is again 0.20 and the Wald rejection rate is 0.728. `recoverBow` means are $b_{12} = 0.493$, $b_{23} = -0.498$, $G_{11} = -0.123$, and $G_{22} = -0.508$, with coverage between 0.920 and 0.955. The sign reading changes the parameter values and does not change the bias or the coverage.

15 Calibration to three phenotypic correlations

This section asks whether three published wheat correlations can stand in for the missing line data. They cannot. The run is a scale check, not a replication of Table 1. Campbell et al., Crop Science 43 (2003), Table 2, print phenotypic correlations for 95 recombinant inbred chromosome lines pooled across seven environments. There are no standard deviations and no environment or block adjustment. *(taken from a cited source, with the limits just stated)* The text layer of the copy we could read dropped minus signs. The paragraph above the table says that SPSM and kernels per square meter were negatively correlated with thousand-kernel weight. We therefore use $\mathrm{Corr}(\mathrm{TKWT}, \mathrm{SPSM}) = -0.44$. The cells $\mathrm{Corr}(\mathrm{TKWT}, \mathrm{KPS}) = 0.08$ and $\mathrm{Corr}(\mathrm{KPS}, \mathrm{SPSM}) = 0.59$ are the signs in that text layer; the prose does not restate them. *(our assumption for those two signs)* GYLD is the fourth trait in the application and is not in equation (7). The three-trait order is $Y_1 = \mathrm{TKWT}$, $Y_2 = \mathrm{KPS}$, $Y_3 = \mathrm{SPSM}$, which is the chain used in the indirect-effect arithmetic on PDF page 17 of Mi et al. *(our assumption)*

Treating the correlation matrix as a covariance, the diagonal recursive factorization is

$$B = \begin{pmatrix} 0 & 0.521 & -0.747 \\ 0 & 0 & 0.590 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \Psi = \mathrm{diag}(0.629, 0.652, 1).$$

(our own calculation) The eigenvalues of the correlation are 0.224, 1.077, and 1.699, so this sign choice is positive definite. Naive Gaussian standard errors that pretend the matrix is a sample correlation of 95 i.i.d. lines are about 0.10 for the two coefficients in the first equation and 0.084 for 0.590. Campbell pooled seven environments, so those standard errors are not his sampling variances. The point fit does not depend on N.

The scan was rerun with this B and Ψ and with the signed G from the position reading of equation (7). Seed `20101029`, 100 replicates. Those QTL coefficients were not estimated from the correlations, and on this residual scale they are large: multi-trait power is 93%, 95%, and 99%. Single-trait power for QTL3 on Y_1 is 96%. That is a consequence of the scale mixture, not evidence that the 2003 correlations produce Mi's powers. *(our own simulation, under an assumption we flag)* The bow check at $\rho = 0.25$ (seed `20101030`) gives a diagonal probability limit 0.198 for the missing edge and a rejection rate 0.74. The unsigned-G run of the same seed had multi-trait powers 100%, 96%, and 100%. The sign change does not turn the calibration into a replication.

16 Limitations

This section states what the proofs and the simulations do not settle. The wheat data fit is blocked, the sampling distribution is not a Lean theorem, and the SEM powers in Table 1 remain unmatched. The wheat fit is still blocked. It needs the 98 line identifiers, the ANOVA residuals of grain yield, thousand-kernel weight, kernels per spike, and spikes per square meter, the ± 1 QTL and cofactor codes, the divisor-n sample covariance, and the edge set as a matrix. Figure 5 draws arrows and does not print that matrix. Campbell's Table 2 does not substitute for any of those columns.

The Drton–Foygel–Sullivant criterion is not a theorem of this repository. The delta method is not a Lean central limit theorem. The position reading of equation (7) meets Table 2's parameter block up to the two rounded QTL2 entries. It leaves the SEM powers for QTL2 and QTL3, and about eleven points of the single-trait QTL3 power, short of Table 1. A second block tested the choices the simulation paragraph does not pin down. None of them closes the QTL2 gap. A 250-line scan matches that one cell and misses the first QTL. Those leftovers are not Monte Carlo error at 1000 replicates. Twelve generations of selfing, Haldane's map, the marker placement, and the detection windows are our assumptions. The 2021 Hidalgo-Contreras wheat SEM is a different experiment.

17 Reproducibility

This section says how to rerun the calculations. The command list for every script is `formalization/correlated-residuals/README.md`. From `formalization/correlated-residuals/simulation`,

```
python3 studies.py --reps 100 --bow-reps 400 --studies 1,2,3
```

writes `study1.json`, `study2.json`, and `study3.json`. The CSV tables are built from those files.

- Simulation commit: `5cd3c9f`.
- Full hash: `5cd3c9f6d2f6e38f25e32146d74d55f3778436e1`.
- Branch: `cursor/l-f102-correlated-residuals-4c01`.
- Lean 4.22.0 and Mathlib v4.22.0, built with `lake build`.
- Theorem links are pinned to `05d69ea141adf3d5ee936d3b48061bde241250dc`.
- This PDF is `pdflatex paper.tex`. The ledger is `SOURCES.md`.
- The gap grid is `python3 gaps.py` in the simulation directory. Seeds are `20101027` through `20101042`.

A Central Lean proofs

This appendix prints the theorem bodies named in Section 1. The files linked there contain the lemmas. The diagonal model reads the structural coefficients from the Schur complement $\Omega = \Sigma_{yy} - \Sigma_{yx}\Sigma_{xx}^{-1}\Sigma_{yx}^{\mathsf{T}}$, then recovers $\Gamma = (I - B)\Sigma_{yx}\Sigma_{xx}^{-1}$ and $\Phi = \Sigma_{xx}$. The bow-free model drops the edge into the first trait from the third and reads the residual covariance c from the same complement. The collision proofs exhibit, for an arbitrary exogenous coefficient, a second parameter point with

the same joint blocks. Consistency is continuous mapping of those left inverses, composed with the strong law for the sample Gram matrix. The listings below are the theorem bodies. Their lemmas live in the linked files.

`recover_sigma`, `Diagonal.lean`, lines 246–279.

```
theorem recover_sigma (θ : Params) (h : OnDomain θ) : recover (sigma θ) = θ := by
  obtain ⟨hv1, hv2, hv3, hφ⟩ := h
  have _hpsi : (psi θ).PosDef := by
    rw [psi, posDef_diagonal_iff]
    intro i
    fin_cases i
    · simpa using hv1
    · simpa using hv2
    · simpa using hv3
  have hread : readDiag (omega θ) = (θ.b12, θ.b13, θ.b23, θ.v1, θ.v2, θ.v3) :=
    readDiag_omega θ hv2 hv3
  have hschur := schur_omega θ hφ
  simp only [recover, hschur, hread]
  have hB : (1 -
      ![0, θ.b12, θ.b13;
        0, 0, θ.b23;
        0, 0, 0]) = subB θ := by
    simpa [Bmat] using (subB_eq_one_sub θ).symm
  have hdet : IsUnit θ.phi.det := (isUnit_iff_isUnit_det θ.phi).1 hφ.isUnit
  apply Params.ext
  · rfl
  · rfl
  · rfl
  · rfl
  · rfl
  · rfl
  · simp only [sigma]
    rw [hB]
    calc
      subB θ * (Lmat θ * θ.gamma * θ.phi) * θ.phi⁻¹
          = (subB θ * Lmat θ) * θ.gamma * (θ.phi * θ.phi⁻¹) := by simp only [mul_assoc]
      _ = (1 : M3) * θ.gamma * (1 : M3) := by rw [subB_mul_L, mul_nonsing_inv _ hdet]
      _ = θ.gamma := by simp
  · rfl
```

`recover_bow`, `BowFree.lean`, lines 187–213.

```
theorem recover_bow (θ : Bow) (h : OnBow θ) : recoverBow θ.sigma = θ := by
  obtain ⟨hN, hφ⟩ := h
  have hv2 := Bow.v2_pos θ hN
  have hv3 := Bow.v3_pos θ hN
  have hread := readBow_omega θ hv2 hv3
  have hschur : θ.sigma.yy - θ.sigma.yx * θ.sigma.xx⁻¹ * θ.sigma.yxᵀ = θ.omega := by
    simpa [Bow.sigma, Bow.omega] using
      schur_noise (L := θ.L) (G := θ.gamma) (Φ := θ.phi) (N := θ.noise) hφ
  simp only [recoverBow, hschur, hread]
  have hB : (1 - ![0, θ.b12, 0; 0, 0, θ.b23; 0, 0, 0]) = θ.subB := by
    simpa [Bow.B] using (Bow.subB_eq θ).symm
  have hdet : IsUnit θ.phi.det := (isUnit_iff_isUnit_det θ.phi).1 hφ.isUnit
  apply Bow.ext
  · rfl
  · rfl
  · rfl
```

```
  · rfl
  · rfl
  · rfl
  · simp only [Bow.sigma]
    rw [hB]
    calc
      θ.subB * (θ.L * θ.gamma * θ.phi) * θ.phi⁻¹
          = (θ.subB * θ.L) * θ.gamma * (θ.phi * θ.phi⁻¹) := by simp only [mul_assoc]
      _ = (1 : M3) * θ.gamma * (1 : M3) := by rw [Bow.subB_mul_L, mul_nonsing_inv _ hdet]
      _ = θ.gamma := by simp
  · rfl
```

`bow12_same_law` in `Collision.lean`, lines 114–128. The siblings `bow13_same_law` and `bow23_same_law` repeat the argument on the other two pairs, lines 130–160.

```
theorem bow12_same_law (c : ℝ) (G Φ : M3) :
    bow12L c * (bow12G c * G) * Φ = G * Φ ∧
      bow12L c * ((bow12G c * G) * Φ * (bow12G c * G)ᵀ + bow12G c * (bow12G c)ᵀ) * (bow12L c)ᵀ =
        G * Φ * Gᵀ + 1 := by
  have hG : bow12L c * (bow12G c * G) = (1 : M3) * G := by
    rw [← mul_assoc, bow12_resolve c, one_mul]
  have hN : bow12L c * (bow12G c * (bow12G c)ᵀ) * (bow12L c)ᵀ = (1 : M3) :=
    resolved_noise (bow12_resolve c)
  have hbaseN : (1 : M3) * (1 : M3) * (1 : M3)ᵀ = (1 : M3) := by simp
  obtain ⟨hyx, hyy⟩ := exogenous_lift (L := bow12L c) (G := bow12G c * G)
    (L' := (1 : M3)) (G' := G) (N := bow12G c * (bow12G c)ᵀ) (N' := (1 : M3)) (Φ := Φ)
    hG (hN.trans hbaseN.symm)
  constructor
  · simpa using hyx
  · simpa using hyy
```

`unitUpper_factor_unique`, `Triangular.lean`, lines 68–135.

```
theorem unitUpper_factor_unique {n : ℕ}
    {L M : Matrix (Fin n) (Fin n) ℝ} {d e : Fin n → ℝ}
    (hL : UnitUpper L) (hM : UnitUpper M)
    (hd : ∀ i, 0 < d i) (he : ∀ i, 0 < e i)
    (hΩ : L * diagonal d * Lᵀ = M * diagonal e * Mᵀ) :
    L = M ∧ d = e := by
  induction n with
  | zero =>
    constructor
    · ext i j
      exact i.elim0
    · ext i
      exact i.elim0
  | succ n ih =>
    have hde_last : d (Fin.last n) = e (Fin.last n) := by
      have hLlast := gram_last (d := d) hL (Fin.last n)
      have hMlast := gram_last (d := e) hM (Fin.last n)
      rw [hL.2] at hLlast
      rw [hM.2] at hMlast
      have hentry := congr_fun (congr_fun hΩ (Fin.last n)) (Fin.last n)
      rw [hLlast, hMlast] at hentry
      simpa using hentry
    have hcol : ∀ i, L i (Fin.last n) = M i (Fin.last n) := by
      intro i
      have hLi := gram_last (d := d) hL i
```

```
    have hMi := gram_last (d := e) hM i
    have hentry := congr_fun (congr_fun hΩ i) (Fin.last n)
    rw [hLi, hMi, hde_last] at hentry
    exact mul_right_cancel₀ (he (Fin.last n)).ne' hentry
  have hlead :
      lead L * diagonal (fun k : Fin n => d k.castSucc) * (lead L)ᵀ =
        lead M * diagonal (fun k : Fin n => e k.castSucc) * (lead M)ᵀ := by
    ext i j
    have hLij := lead_gram (L := L) (d := d) i j
    have hMij := lead_gram (L := M) (d := e) i j
    have hentry := congr_fun (congr_fun hΩ i.castSucc) j.castSucc
    rw [hLij, hMij] at hentry
    have hr :
        L i.castSucc (Fin.last n) * d (Fin.last n) * L j.castSucc (Fin.last n) =
          M i.castSucc (Fin.last n) * e (Fin.last n) * M j.castSucc (Fin.last n) := by
      rw [hcol, hcol, hde_last]
    rw [hr] at hentry
    exact add_right_cancel hentry
  have hrec :=
    ih (lead_unitUpper hL) (lead_unitUpper hM) (fun i => hd i.castSucc)
      (fun i => he i.castSucc) hlead
  refine ⟨?_, ?_⟩
  · ext i j
    cases i using Fin.lastCases with
    | last =>
      cases j using Fin.lastCases with
      | last =>
        rw [hL.2, hM.2]
      | cast j =>
        rw [hL.1 _ _ j.castSucc_lt_last, hM.1 _ _ j.castSucc_lt_last]
    | cast i =>
      cases j using Fin.lastCases with
      | last =>
        exact hcol i.castSucc
      | cast j =>
        have hlead_ij := congr_fun (congr_fun hrec.1 i) j
        simpa [lead, submatrix_apply] using hlead_ij
  · ext i
    cases i using Fin.lastCases with
    | last => exact hde_last
    | cast i =>
      have := congr_fun hrec.2 i
      simpa using this
```

sigmaRec_injective, Triangular.lean, lines 298–317.

```
theorem sigmaRec_injective {n p : ℕ} {θ θ' : RecModel n p}
    (hθ : OnRec θ) (hθ' : OnRec θ') (h : sigmaRec θ = sigmaRec θ') : θ = θ' := by
  have hθ0 := hθ
  have hθ'0 := hθ'
  obtain ⟨hB, hd, hΦ⟩ := hθ
  obtain ⟨hC, he, hΦ'⟩ := hθ'
  have hgram :
      (1 - θ.B)⁻¹ * diagonal θ.d * ((1 - θ.B)⁻¹)ᵀ =
        (1 - θ'.B)⁻¹ * diagonal θ'.d * ((1 - θ'.B)⁻¹)ᵀ := by
    have hθs := schur_sigmaRec θ hΦ
    have hθs' := schur_sigmaRec θ' hΦ'
    rw [h] at hθs
    exact hθs.symm.trans hθs'
```

```
  obtain ⟨hBeq, hdeq⟩ := recursive_residual_unique hB hC hd he hgram
  have hγ : θ.gamma = θ'.gamma := by
    have hg := (recover_exogenous θ hθ0).2
    have hg' := (recover_exogenous θ' hθ'0).2
    rw [hg, hg', h, hBeq]
  have hφ : θ.phi = θ'.phi := by simpa [sigmaRec] using congrArg Joint.xx h
  exact RecModel.ext hBeq hdeq hγ hφ
```

plug_in_consistent_ae, Law.lean, lines 174–199.

```
theorem plug_in_consistent_ae {Ω : Type*} [MeasurableSpace Ω] {μ : Measure Ω}
    {Z : ℕ → Ω → Idx → ℝ} {θ : Params} (hθ : OnDomain θ)
    (hindep : Pairwise fun i j => IndepFun (Z i) (Z j) μ)
    (hident : ∀ i, IdentDistrib (Z i) (Z 0) μ μ)
    (hint : ∀ a b : Idx, Integrable (fun ω => Z 0 ω a * Z 0 ω b) μ)
    (hyy : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inl j) ∂μ = (sigma θ).yy i j)
    (hyx : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inr j) ∂μ = (sigma θ).yx i j)
    (hxx : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inr i) * Z 0 ω (.inr j) ∂μ = (sigma θ).xx i j) :
    ∀ᵐ ω ∂μ, Tendsto (fun n =>
        ((recover (sampleCov Z n ω)).b12, (recover (sampleCov Z n ω)).b13,
          (recover (sampleCov Z n ω)).b23, (recover (sampleCov Z n ω)).v1,
          (recover (sampleCov Z n ω)).v2, (recover (sampleCov Z n ω)).v3,
          (recover (sampleCov Z n ω)).gamma, (recover (sampleCov Z n ω)).phi))
      atTop
      (nhds (θ.b12, θ.b13, θ.b23, θ.v1, θ.v2, θ.v3, θ.gamma, θ.phi)) := by
  filter_upwards [strong_gram Z hindep hident hint] with ω hω
  have hyy' : Tendsto (fun n => (sampleCov Z n ω).yy) atTop (nhds (sigma θ).yy) := by
    simpa [sampleCov, blockYY_pop hyy] using tendsto_blockYY hω
  have hyx' : Tendsto (fun n => (sampleCov Z n ω).yx) atTop (nhds (sigma θ).yx) := by
    simpa [sampleCov, blockYX_pop hyx] using tendsto_blockYX hω
  have hxx' : Tendsto (fun n => (sampleCov Z n ω).xx) atTop (nhds (sigma θ).xx) := by
    simpa [sampleCov, blockXX_pop hxx] using tendsto_blockXX hω
  exact plug_in_consistent hθ hyy' hyx' hxx'
```

plug_in_bow_consistent_ae, BowConsistency.lean, lines 270–304.

```
theorem plug_in_bow_consistent_ae {Ω : Type*} [MeasurableSpace Ω] {μ : Measure Ω}
    {Z : ℕ → Ω → Idx → ℝ} {θ : Bow} (hθ : OnBow θ)
    (hindep : Pairwise fun i j => IndepFun (Z i) (Z j) μ)
    (hident : ∀ i, IdentDistrib (Z i) (Z 0) μ μ)
    (hint : ∀ a b : Idx, Integrable (fun ω => Z 0 ω a * Z 0 ω b) μ)
    (hyy : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inl j) ∂μ = θ.sigma.yy i j)
    (hyx : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inl i) * Z 0 ω (.inr j) ∂μ = θ.sigma.yx i j)
    (hxx : ∀ i j : Fin 3,
      ∫ ω, Z 0 ω (.inr i) * Z 0 ω (.inr j) ∂μ = θ.sigma.xx i j) :
    ∀ᵐ ω ∂μ, Tendsto (fun n =>
        ((recoverBow (sampleCov Z n ω)).b12, (recoverBow (sampleCov Z n ω)).b23,
          (recoverBow (sampleCov Z n ω)).v1, (recoverBow (sampleCov Z n ω)).v2,
          (recoverBow (sampleCov Z n ω)).v3, (recoverBow (sampleCov Z n ω)).c,
          (recoverBow (sampleCov Z n ω)).gamma, (recoverBow (sampleCov Z n ω)).phi))
      atTop
      (nhds (θ.b12, θ.b23, θ.v1, θ.v2, θ.v3, θ.c, θ.gamma, θ.phi)) := by
  filter_upwards [strong_gram Z hindep hident hint] with ω hω
  have hyy' : Tendsto (fun n => (sampleCov Z n ω).yy) atTop (nhds θ.sigma.yy) := by
```

```
  have hpop : blockYY (popGram μ (Z 0)) = θ.sigma.yy := by
    ext i j
    simpa [blockYY, popGram] using hyy i j
  simpa [sampleCov, hpop] using tendsto_blockYY hω
have hyx' : Tendsto (fun n => (sampleCov Z n ω).yx) atTop (nhds θ.sigma.yx) := by
  have hpop : blockYX (popGram μ (Z 0)) = θ.sigma.yx := by
    ext i j
    simpa [blockYX, popGram] using hyx i j
  simpa [sampleCov, hpop] using tendsto_blockYX hω
have hxx' : Tendsto (fun n => (sampleCov Z n ω).xx) atTop (nhds θ.sigma.xx) := by
  have hpop : blockXX (popGram μ (Z 0)) = θ.sigma.xx := by
    ext i j
    simpa [blockXX, popGram] using hxx i j
  simpa [sampleCov, hpop] using tendsto_blockXX hω
exact plug_in_bow_consistent hθ hyy' hyx' hxx'
```

B Artifacts

This appendix maps each claim to the file and the command that regenerate it. Each row is one claim in this note, the file that holds it, and the command that regenerates it. Commands are from the repository root unless the row says otherwise. Commits are on `cursor/l-f102-correlated-residuals-4c01`. A short `-reps` run overwrites the committed JSON; the commands below are the runs that match the cited files.

Claim	File and commit	Command
Minus-sign positions in equation (7): equals ends at 145.9, parenthesis at 148.1, four unary minuses	`simulation/out/qtl1/glyphs.json`, `b4e8a3c594f12f419560d5cb2a569b649ae93a55`	`python3formalization/correlated-residuals/simulation/measure_glyphs.py--pdflocal-extraction/literature/unl-sem/mi-qtl2010.pdf--checkformalization/correlated-residuals/simulation/out/qtl1/glyphs.json`
Total effects, including 3/4 and −7/16	`simulation/out/table2_totals.json`	`python3formalization/correlated-residuals/simulation/table2_totals.py`
Baseline SEM powers 50.8, 53.5, 86.1	`simulation/out/assumptions/baseline.json`, `82ff59d6bcfb02c273aaf1f83195e0d120fe0915`	`python3assumptions.py--configsbaseline--reps1000` in `simulation/`
Gap grid, including the 13% and 86.2% single-trait cells	`simulation/out/gaps.json`, `gaps_table.csv`, `c2769042e51806a77b2b70237fb4c014552e78fb`	`python3gaps.py` in `simulation/`
Closest triple 65.2, 73.3, 99.3	`simulation/out/qtl1/qtl1_24_n250.json`, `b4e8a3c594f12f419560d5cb2a569b649ae93a55`	`python3qtl1_search.py--configsqtl1_24_n250--reps1000` in `simulation/`
Product 1.37×10^{-12}	`simulation/out/printed_probability.json`, from the baseline file above	`python3formalization/correlated-residuals/simulation/printed_probability.py`

Claim	File and commit	Command
Studies 1–3, bow Monte Carlo, Campbell scale check	`simulation/out/study1.json`, `study2.json`, `study3.json`, `5cd3c9f6d2f6e38f25e32146d74d55f3778436e1`	`python3studies.py--reps100--bow-reps400--studies1,2,3` in `simulation/`
Identification algebra at the earlier sign convention	`witnesses.json`, written by `check_witness.py`	`python3check_witness.py` in `formalization/correlated-residuals/`
Delta-method numeric check	`delta_check.json`	`python3delta_method.py` in `formalization/correlated-residuals/`
Bow-free plug-in numeric check	`fit_bow.json`	`python3fit_bow.py` in `formalization/correlated-residuals/`
Finished Lean proofs. Axioms: propositional extensionality, choice, and the construction of the reals	`CorrelatedResiduals/*.lean`, listings pinned to `05d69ea141adf3d5ee936d3b48061bde241250dc`	`lakebuild` in `formalization/correlated-residuals/`
This PDF	`paper/paper.pdf`	`pdflatexpaper.tex` in `paper/`

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